BIOSTAT 702: Exercise 5

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Question 1

```r
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to bec
```

```r
data_ex5 <-  read.csv("exercise5-1.csv")

# Set the reference group for later ANOVA modeling
data_ex5$group <- factor(data_ex5$group, levels = c("C", "A", "B"))

# Check the data structure, first rows, group counts, and score range
str(data_ex5)
```

```
## 'data.frame':    150 obs. of  2 variables:
##  $ group: Factor w/ 3 levels "C","A","B": 2 2 2 2 2 2 2 2 2 2 ...
##  $ y    : num  66.1 63.5 44.5 61.7 66.2 ...
```

```r
head(data_ex5)
```

```
##   group     y
## 1     A 66.09
## 2     A 63.46
## 3     A 44.49
## 4     A 61.74
## 5     A 66.23
## 6     A 52.22
```

```
table(data_ex5$group)
```

```
##
##  C  A  B
## 50 50 50
```

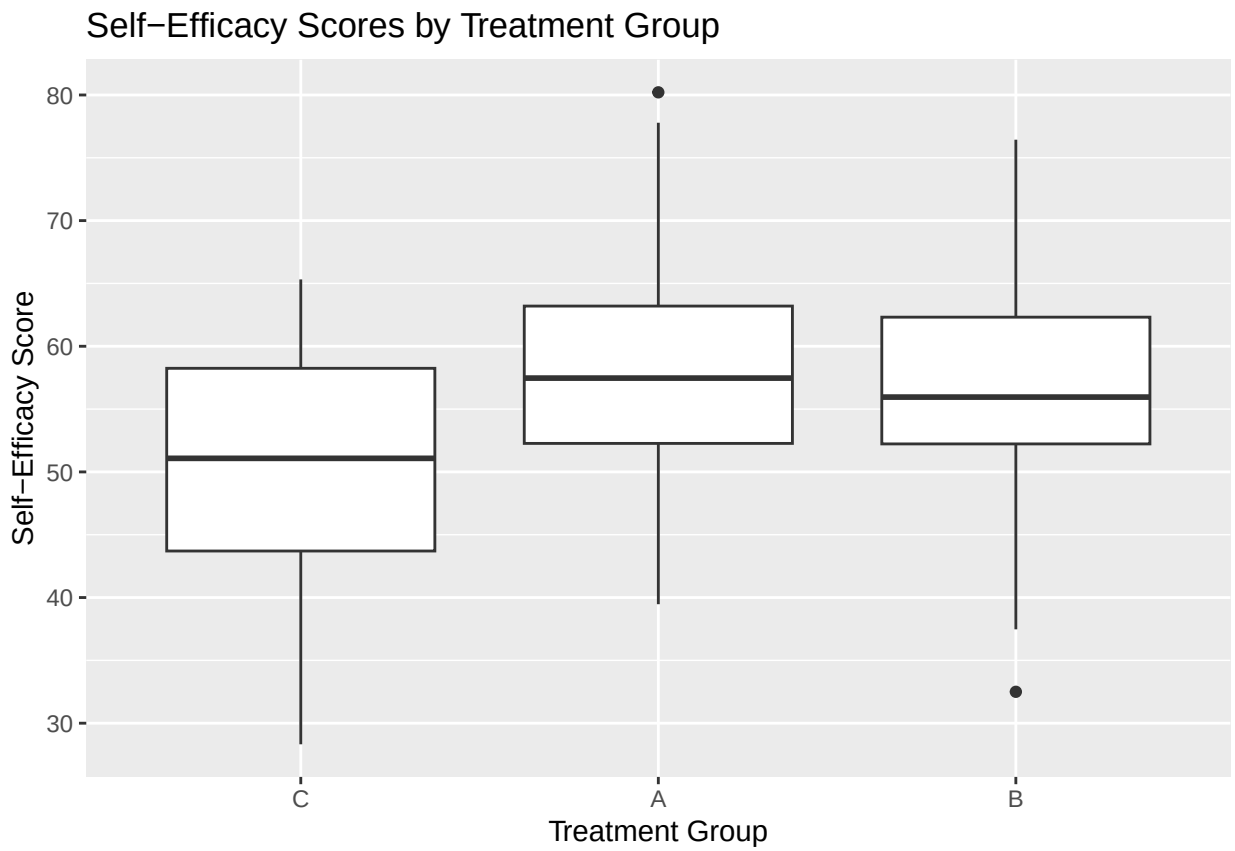
```
range(data_ex5$y)
```

```
## [1] 28.32 80.21
```

Question 1.1

Perform a visualization exercise by creating histograms, boxplots or similar 1-variable visual summaries for each group separately. You're checking for impossible data values, extreme outliers and extreme skewness. You also want the standard deviations to be similar across group, although the results will typically be robust to small to moderate departures from this assumption. Is it appropriate to proceed with an ANOVA?

```
# Create a boxplot to compare scores across treatment groups
ggplot(data = data_ex5, aes(x = group, y = y)) +
  geom_boxplot() +
  labs(
    title = "Self-Efficacy Scores by Treatment Group",
    x = "Treatment Group",
    y = "Self-Efficacy Score"
  )
```




```
# Create a histogram for each treatment group
ggplot(data = data_ex5, aes(x = y)) +
  geom_histogram(bins = 10) +
  facet_wrap(~ group) +
  labs(
    title = "Distribution of Self-Efficacy Scores by Treatment Group",
    x = "Self-Efficacy Score",
    y = "Count"
  )
```



ANSWER The scores were within the possible range of 0 to 100. The histograms do not show severe skewness, and the spreads appear reasonably similar across groups. The boxplots flag a potential high outlier in Group A and a potential low outlier in Group B, but neither appears extreme. Overall, it seems appropriate to proceed with a one-way ANOVA.

Question 1.2

Calculate the group means and standard deviations.

```
# Calculate the mean and SD

data_ex5 %>%
  group_by(group) %>%
  summarise(
    n = n(),
    mean_y = mean(y),
    sd_y = sd(y)
  )
```

```
## # A tibble: 3 x 4
##   group     n mean_y  sd_y
##   <fct> <int>  <dbl> <dbl>
```

```
## 1 C        50   50.7  9.44
## 2 A        50   57.8  9.25
## 3 B        50   56.1  9.19
```

ANSWER The sample mean self-efficacy scores were 57.77 in Group A, 56.14 in Group B, and 50.74 in Group C. The corresponding standard deviations were 9.25, 9.19, and 9.44, respectively. The standard deviations are similar across the three groups, suggesting comparable variability.

Question 1.3

Report the ANOVA table, including an overall F-test, assessing the null hypothesis that the means of all 3 groups are identical (i.e., and thus the observed differences between group means are entirely due to chance). What is the p-value for this?

```
fit1 <- lm(y ~ group, data = data_ex5)
anova(fit1)
```

```
## Analysis of Variance Table
##
## Response: y
##            Df  Sum Sq Mean Sq F value   Pr(>F)
## group       2  1353.7  676.86  7.8339 0.000585 ***
## Residuals 147 12700.9   86.40
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANSWER The overall ANOVA was statistically significant, $F(2, 147) = 7.8339$, $p = 0.000585$. We reject the null hypothesis that the mean self-efficacy scores are equal across all three groups. This provides evidence that at least one group mean differs; the overall test does not identify which groups differ.

Question 1.4

Explain what each of the elements of the ANOVA table mean in this context (e.g., the DF, Sums of Squares, Mean Square, F, and p-value?).

ANOVA element	Interpretation
Df for group (2)	The between-group degrees of freedom are $3 - 1 = 2$ because there are three treatment groups.
Df for residuals (147)	The residual degrees of freedom are 150 participants $-$ 3 groups $= 147$.
Sum of Squares for group (1,353.7)	This measures the variation in self-efficacy scores associated with differences among the three group means.

ANOVA element	Interpretation
Sum of Squares for residuals (12,700.9)	This measures the variation in participants' scores within their respective treatment groups.
Mean Square for group (676.86)	This is the group sum of squares divided by its degrees of freedom.
Mean Square for residuals (86.40)	This is the residual sum of squares divided by its degrees of freedom and estimates the within-group error variance.
F value (7.8339)	This is the ratio of the group mean square to the residual mean square: 676.86 / 86.40.
p-value (0.000585)	Assuming all three population means are equal, this is the probability of observing an F statistic of 7.8339 or larger.

Question 2

The investigator proposes 4 hypotheses of scientific interest.

Question 2.1

For practice, we will perform "unadjusted" and "adjusted" analyses, where the unadjusted analyses use a test-wise type 1 error rate and the adjusted analyses use an experiment-wide type 1 error rate. In plain English, what is the difference between the two? In practice, you should decide which approach to use before starting data analysis.

ANSWER

Unadjusted analyses control the Type I error rate for each individual test. Adjusted analyses account for the full set of planned tests and control the probability of making at least one Type I error across that set. The analysis approach should be chosen before examining the results.

Question 2.2

The investigator proposes 4 hypotheses of scientific interest. 1. The mean of group A differs from group B. 2. The mean of group A differs from group C. 3. The mean of group B differs from group C. 4. The average mean of two intervention groups, A and B, differ from the control, C. For each of the hypotheses, state the contrast that is associated with that hypothesis. Remember how your group variable is coded and what each coefficient actually represents.

```
L <-  rbind(
  c(0, 1, -1),     # H1
  c(0, 1, 0),      # H2
  c(0, 0, 1),      # H3
  c(0, 1/2, 1/2)   # H4
)
```

```
L
```

```
##      [,1] [,2] [,3]
## [1,]    0  1.0 -1.0
## [2,]    0  1.0  0.0
## [3,]    0  0.0  1.0
## [4,]    0  0.5  0.5
```

ANSWER

Hypothesis	Contrast in group means	Contrast vector
H1: Group A differs from Group B	$\mu_A - \mu_B$	(0, 1, −1)
H2: Group A differs from Group C	$\mu_A - \mu_C$	(0, 1, 0)
H3: Group B differs from Group C	$\mu_B - \mu_C$	(0, 0, 1)
H4: The average of Groups A and B differs from Group C	$(\mu_A + \mu_B)/2 - \mu_C$	(0, 1/2, 1/2)

With Group C as the reference group, the intercept represents the mean for Group C. The `groupA` and `groupB` coefficients represent the differences between Groups A and C, and Groups B and C, respectively.

Question 2.3

Using an unadjusted approach, for each hypothesis provide a point estimate, confidence interval, and p-value.

```
contrast_unadjusted <- doBy::esticon(
  fit1, L, conf.int = TRUE, level = 0.95) %>%
  as.data.frame() %>%
  select(estimate, lwr, upr, p.value)

contrast_unadjusted
```

```
##   estimate       lwr       upr      p.value
## 1   1.6274 -2.046500  5.301300 0.3827859209
## 2   7.0286  3.354700 10.702500 0.0002267588
## 3   5.4012  1.727300  9.075100 0.0042355693
## 4   6.2149  3.033209  9.396591 0.0001691727
```

ANSWER

Hypothesis	Estimate	95% confidence interval	p-value
H1: A vs. B	1.6274	(−2.0465, 5.3013)	0.3828
H2: A vs. C	7.0286	(3.3547, 10.7025)	0.0002
H3: B vs. C	5.4012	(1.7273, 9.0751)	0.0042
H4: Average of A and B vs. C	6.2149	(3.0332, 9.3966)	0.0002

Using an unadjusted significance level 0.05, the comparisons of Groups A and C, Groups B and C, and the average of Group A and B with Group C were statistically significant. The comparison of Groups A and B was not statistically significant; its confidence interval includes zero.

Question 2.4

Using Bonferroni adjustment, for each hypothesis provide a point estimate, confidence interval, and p-value. (Some packages literally change the p-value, whereas others keep the p-value and change the threshold for declaring statistical significance.)

```
contrasts2 <- doBy::esticon(
  fit1,
  L,
  conf.int = TRUE,
  level = 1 - 0.05 / dim(L)[1]
) %>%
  as.data.frame()

# Adjust the p-values using the Bonferroni method
contrasts2$p.value <- p.adjust(
  contrasts2$p.value,
  method = "bonferroni"
)

# Display the estimates, adjusted confidence intervals, and adjusted p-values
contrasts2 %>%
  select(estimate, lwr, upr, p.value)
```

```
##   estimate        lwr       upr      p.value
## 1   1.6274 -3.0737719  6.328572 1.0000000000
## 2   7.0286  2.3274281 11.729772 0.0009070351
## 3   5.4012  0.7000281 10.102372 0.0169422771
## 4   6.2149  2.1435657 10.286234 0.0006766908
```

ANSWER

Hypothesis	Estimate	98.75% Bonferroni-adjusted CI	Adjusted p-value
H1: A vs. B	1.6274	(−3.0738, 6.3286)	1.0000
H2: A vs. C	7.0286	(2.3274, 11.7298)	0.0009
H3: B vs. C	5.4012	(0.7000, 10.1024)	0.0169
H4: Average of A and B vs. C	6.2149	(2.1436, 10.2862)	0.0007

After the Bonferroni adjustment, H1 was not statistically significant because its confidence interval includes zero and its adjusted p-value is greater than 0.05. H2, H3, and H4 were statistically significant because their confidence intervals exclude zero and their adjusted p-values are below 0.05.

Question 3

Question 3.1

Summarize the results from the ANOVA, using language that a clinical investigator could understand. When describing the Question 2 results, focus on the multiple-test-corrected ones.

```
# Display the overall ANOVA results
anova(fit1)
```

```
## Analysis of Variance Table
##
## Response: y
##            Df  Sum Sq Mean Sq F value   Pr(>F)
## group       2  1353.7  676.86  7.8339 0.000585 ***
## Residuals 147 12700.9   86.40
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Display the Bonferroni-adjusted contrast results
contrasts2 %>%
  select(estimate, lwr, upr, p.value)
```

```
##   estimate        lwr       upr      p.value
## 1   1.6274 -3.0737719  6.328572 1.0000000000
## 2   7.0286  2.3274281 11.729772 0.0009070351
## 3   5.4012  0.7000281 10.102372 0.0169422771
## 4   6.2149  2.1435657 10.286234 0.0006766908
```

ANSWER At 6 months, mean self-efficacy scores differed across the three groups, as indicated by the one-way ANOVA, $F(2, 147) = 7.83$, $p < 0.001$. The mean scores were 57.77 for Group A, 56.14 for Group B, and 50.74 for Group C. After Bonferroni adjustment, Group A had a higher mean score than Group C by 7.03 points (98.75% CI: 2.33 to 11.73, adjusted $p = 0.0009$), and Group B had a higher mean score than Group C by 5.40 points (98.75% CI: 0.70 to 10.10, adjusted $p = 0.0169$). The average of Groups A and B was 6.21 points higher than Group C (98.75% CI: 2.14 to 10.29, adjusted $p = 0.0007$). The difference between Groups A and B was not statistically significant after adjustment (estimate = 1.63 points, 98.75% CI: −3.07 to 6.33, adjusted $p = 1.0000$).

Question 3.2

As a small subtlety, comparing groups A and B within the context of a 1-way ANOVA will provide a different result than ignoring group C and performing a t-test. Why?

```
# ANOVA
contrast_ab <- doBy::esticon(
  fit1,
  L[1, , drop = FALSE],
  conf.int = TRUE
) %>%
  as.data.frame()

contrast_ab %>%
  select(estimate, std.error, p.value)
```

```
##   estimate std.error   p.value
## 1   1.6274  1.859042 0.3827859
```

```
# T-test
data_ab <- data_ex5 %>%
  filter(group %in% c("A", "B")) %>%
  droplevels()

t_test_ab <- t.test(
  y ~ group,
  data = data_ab,
  var.equal = TRUE
)

t_test_ab
```

```
##
##  Two Sample t-test
##
## data:  y by group
## t = 0.88242, df = 98, p-value = 0.3797
```

```
## alternative hypothesis: true difference in means between group A and group B is not equal to
## 95 percent confidence interval:
##  -2.032428  5.287228
## sample estimates:
## mean in group A mean in group B
##         57.7678         56.1404
```

ANSWER

The estimated difference in mean self-efficacy scores between Groups A and B was 1.63 points in both analyses. The unadjusted comparison within the one-way ANOVA gave a 95% confidence interval of (−2.05, 5.30) and a p-value of 0.382. The two-sample t-test using only Groups A and B gave a 95% confidence interval of (−2.03, 5.29) and a p-value of 0.380. Neither analysis found a statistically significant difference. The results differ slightly because the one-way ANOVA estimates the pooled within-group variance using all three groups, whereas the two-sample t-test uses only Groups A and B; their degrees of freedom also differ.

Question 4

Now suppose the investigators are only interested in the three pairwise comparisons (i.e., Hypotheses 1-3 from Question 2).

Question 4.1

Recalculate the Bonferroni-adjusted p-values now that the number of hypotheses has changed from Q2. Did your conclusions change?

```r
# Calculate Bonferroni-adjusted p-values for the three pairwise comparisons
pairs(
  emmeans::emmeans(fit1, ~ group),
  adjust = "bonferroni"
)
```

```
##  contrast estimate   SE  df t.ratio p.value
##  C - A       -7.03 1.86 147  -3.781  0.0007
##  C - B       -5.40 1.86 147  -2.905  0.0127
##  A - B        1.63 1.86 147   0.875  1.0000
##
## P value adjustment: bonferroni method for 3 tests
```

ANSWER After applying the Bonferroni adjustment for three pairwise comparisons, the adjusted p-values were 0.0007 for Groups A and C, 0.0127 for Groups B and C, and 1.0000 for Groups A and B. The differences between Groups A and C and between Groups B and C were statistically significant, while the difference between Groups A and B was not. These conclusions are the same as in Question 2, although the adjusted p-values differ because the Bonferroni adjustment now accounts for three hypotheses instead of four.

Question 4.2

Calculate the Tukey HSD adjusted p-values. How do the results compare to the Bonferroni procedure? Is this procedure more or less conservative (i.e., conservative means it's harder to declare a statistically significant result)?

```
# Calculate Tukey HSD-adjusted p-values for all pairwise comparisons
pairs(
  emmeans::emmeans(fit1, ~ group),
  adjust = "tukey"
)
```

```
##  contrast estimate   SE  df t.ratio p.value
##  C - A       -7.03 1.86 147  -3.781  0.0007
##  C - B       -5.40 1.86 147  -2.905  0.0117
##  A - B        1.63 1.86 147   0.875  0.6566
##
## P value adjustment: tukey method for comparing a family of 3 estimates
```

ANSWER The conclusions were the same under both procedures: the differences between Groups A and C and between Groups B and C were statistically significant, while the difference between Groups A and B was not. The Tukey-adjusted p-values were equal to or smaller than the Bonferroni-adjusted p-values, so Tukey HSD was less conservative for these comparisons.

Question 5

Please answer the following questions regarding the FDA guidance on Multiple Testing

Question 5.1

What is the main statistical concern when analyzing multiple endpoints in a clinical trial?

ANSWER The main statistical concern is inflation of the Type I error rate. When multiple endpoints are analyzed and a treatment is considered effective if any one endpoint is statistically significant, testing each endpoint without an appropriate multiplicity strategy increases the chance of a false-positive conclusion. This can lead to erroneous or misleading claims about the treatment's effects.

Question 5.2

Summarize the FDA's recommendations for handling multiplicity in clinical trials.

ANSWER The FDA recommends planning for multiplicity before the trial begins. Sponsors should prespecify the primary and secondary endpoint families, all planned analyses, and a testing strategy that controls the overall Type I error rate. Secondary endpoints intended to support claims of additional treatment effects should be included in this error-control plan, while exploratory

endpoints generally do not require multiplicity adjustment when they are not used to support formal conclusions. The selected testing strategy should also be considered when planning the sample size to maintain adequate power. The analysis plan should be established before treatment assignments are unmasked; adding analyses after reviewing the results can reintroduce multiplicity concerns.

Question 5.3

What should be considered when dealing with multiple primary endpoints, particularly regarding how success is defined across those endpoints?

ANSWER The trial's success criteria should be clearly defined in advance. If success on any one of several primary endpoints is sufficient to conclude that the treatment is effective, the multiple opportunities for success require control of the overall Type I error rate. If success requires demonstrating an effect on all primary endpoints, there is only one path to trial success, so multiplicity-related Type I error inflation is not a concern for those endpoints. However, requiring success on every endpoint can reduce power and increase the risk of failing to demonstrate a real treatment effect. The number of co-primary endpoints and the sample size should therefore be considered carefully.

Question 5.4

What is the statistical impact of controlling for multiple testing?

ANSWER Controlling for multiple testing limits the overall Type I error rate, reducing the chance of falsely concluding that a treatment has an effect on at least one endpoint. However, for a fixed sample size, the more stringent testing criteria can reduce power and increase the risk of a Type II error. The sample size may need to be increased to maintain the desired power.